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Navigating the lunar frontier: one hundred landing sites at the south pole for future mission challenges

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ABSTRACT

Despite humanity's many lunar missions to the equatorial and mid-latitude regions, the south pole remains uncharted because of its exceptionally harsh conditions. The quest for water ice and the drive to establish lunar bases have positioned the south pole area above 80° latitude, characterized by permanently shaded regions and conducive to water ice preservation. However, the daunting terrain and intricate illumination in this area present significant challenges to engineering safety. Here, we introduce a Landing Feasibility Probability (LFP) model to evaluate the viability of potential landing sites. We pinpoint 120 prospective landing sites, stratified into 25 high-priority, 64 medium-priority, and 31 low-priority sites. These sites, encompassing a mere 0.6% of the lunar south pole, have been rigorously vetted against 10 critical factors for landing feasibility. These sites show a pronounced clustering around 8 major craters and plateaus, organizing into 7 lunar site networks, with each network comprising 4 sites with a maximum dimension under 25 km, ideal for the development of lunar bases and observational networks. The LFP model's selection process is derived from a heuristic Genetic Algorithm (GA) informed by expert experience. The landing sites are strategically positioned to address the dual challenges of scientific goals and engineering safety at the lunar south pole, while also providing site selection guidance and facilitating international collaboration and communication for lunar expeditions. This method can also be adapted for site selection on other celestial bodies (e.g. Mars and asteroids) for scientific exploration and construction of extraterrestrial bases.

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1. Introduction

Lunar exploration offers an unparalleled window into the formation and evolution of the solar system and the broader cosmos, while concurrently driving advancements in human technological capabilities [1–3]. In the past 60 years, a total of 24 successful missions have touched down on the Moon's surface, a collaborative achievement by the United States, China, Russia (formerly the USSR), the European Space Agency (ESA), India, and Japan [4–9].

These missions have included landings on both the near and far side of the Moon, as well as sample return operations [10]. However, solely relying on remote sensing and sample collection from the equatorial and mid-latitude regions has its limitations in pushing the boundaries of lunar science. To truly unravel the Moon's evolutionary history and investigate potential biosignatures, missions must target the lunar south pole, a region yet untouched by international exploration efforts [11–14] (Fig. S1 online). As we prepare for this next frontier, identifying viable landing sites at the south pole becomes a critical step [15]. This endeavor presents a confluence of dual challenges of scientific goals and engineering feasibility [16], requiring innovative approaches to

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overcome the obstacles of landing and operating in this uncharted and permanently shadowed lunar environment.

In 2023, India's Chandrayaan-3 made a significant stride in lunar exploration by landing at the latitude of 69 degrees south, nearer to the lunar south pole than any previous international mission [17]. This milestone marks just the beginning of humanity's ventures into this region, with plans for future lunar bases at the South Pole. The USA ambitious Artemis III program [18] is poised to send astronauts to the Moon's South Pole in November 2026, and China's Chang'e-7 mission [19] is slated for a 2026 landing. The lunar south pole holds immense scientific potential due to its unique resources, such as water ice, and its potential to shed light on lunar geology, mineralogy, seismology, and even astronomical observations from a stable lunar platform [20–26]. However, the area's extreme topography and the variability in solar illumination pose significant challenges for detector landing [27–30]. Ideal site selection involves a detailed assessment to ensure a balance between scientific value and engineering safety [31,32]. The landing site size requirement needs to be decided specifically based on the specific mission scenario and planned path length, with some papers including a maximum detection radius of 2 km for astronauts as a screening condition [33]. Selecting optimal sites for lunar missions requires a nuanced understanding of the scientific goals and the engineering feasibility [34,35], and taking into account insights from domain experts to support the continued exploration of the lunar south pole [36,37].

Regarding the site selection methods, the expert working teams need to conduct long-term discussions and assessments during the landing site selection, benefitting the success of the exploration mission [38]. In the NASA Artemis Program, 13 candidate pre-selected landing areas were selected by the expert team considering numerous engineering and scientific factors. For China's Chang'e missions, the specialists have also carried out conference seminars and scientific research, ranging from reviewing the historical site selection to pre-studying the future site selection, contributing greatly to the successes of the landing and roving missions [39,40]. However, the extreme environment at the lunar south pole increases the time cost and makes it difficult for experts to reach an agreement. The reasons behind this include the limited availability of high-quality remote sensing data at the lunar south pole, the uncertainty in qualitative estimation of scientific value, and the subjectively determined rules for site selection.

Visual interpretation based on expert experience is the classic site selection method, combining factors like the Digital Elevation Models (DEMs), the slope, the illumination, and the Earth communication to effectively identify safe landing sites. The previous studies have made efforts to assess and combine diverse engineering factors, guarantee the safe touchdown and rover mobility, while ensuring the solar energy availability [41–45]. Furthermore, the literature often employs the empirical thresholds or Analytic Hierarchy Process (AHP) techniques to define the site selection rules [46]. AHP assigns weight values by judging the relative importance of two factors [47]. Due to the different backgrounds of the experts, a large site selection workgroup can produce multiple combinations of factor weights oriented to different task scenarios, increasing the time cost and difficulty of site selection.

Several studies using learning-based approaches for site selection have recently emerged to explore new approaches to factor weight assignment [48–51]. The Convolutional Neural Network Approach (1D-CNNs) has been employed to quantitatively assess potential landing sites for exploration and lunar base construction, considering both scientific and engineering criteria [15]. However, the internal decision-making process of CNNs remains a black box, with the model parameter optimization process being opaque and difficult to interpret. This lack of transparency may introduce

uncertainties in landing site evaluations, as the critical factors influencing predictions are not explicitly revealed. Therefore, it is necessary to further investigate the mechanisms of learning-based selection methods to improve the reliability and interpretability of lunar exploration missions. The method proposed in this paper aims to model expert site selection experience using a machine learning approach to overcome the weaknesses of the deep learning modeling process, which is opaque and difficult to regulate. More importantly, it provides a new approach for the site selection expert group to complement classical methods such as visual interpretation and AHP to depict the lunar south pole landing site map oriented to water ice exploration.

Addressing the dual challenges of selecting a suitable site at the lunar south pole, we developed an innovative cross-regional heuristic collaborative determination approach (Materials and Methods). This approach synergizes the Engineering Safety Index (ESI) with the Scientific Exploration Value Index (EVI), evolving the site selection process from the over-reliance on expert experience to intelligent discernment. Our methodology prioritizes key scientific factors, such as the presence of water ice, geological diversity, and mineral resources, with particular emphasis on proximity to water ice reservoirs, Permanently Shadowed Regions (PSRs), and cold traps. For engineering safety, we focus on surface slope, average illumination, regolith temperature, flat area size, and communication potential as primary considerations. We employed a genetic algorithm (GA) on 23 site selection factors, incorporating root-mean-square-error (RMSE) as the objective function, to pre-assess the importance of the factors to the site selection task. In the second GA training progress, 13 of the most significant site selection factors were employed to construct a Landing Feasibility Probability (LFP) model. This model is designed to closely emulate the knowledge rules and judgment of domain experts in site selection. For model training and validation, we used positive samples (52,378 pixels with a 20 m spatial resolution), identified by both the USA Artemis III program and domain experts as viable landing sites, and negative samples (518,641 pixels), marked as hazardous by experts. We allocated 70% of these samples to the training dataset and 30% to the validation dataset. Our evaluation metrics exceeded 94.59% impressively in F1 score in correctly identifying positive samples within the training dataset and 90.49% for the validation dataset. This translates to a mere 9.51% probability of model error in site selection, marking the first precision assessment for an extraterrestrial object detection site selection model of its kind. The high accuracy and robustness of the model suggest that it can be a highly reliable tool in the search for lunar south pole exploration landing sites.

2. Materials and methods

2.1. Selection criterion

The selection criteria for potential lunar landing sites are closely tied to the specific goals of exploration missions. In 1970, NASA developed the mission design matrix for the APOLLO MISSION to define the mission comprehensive requirements for sites, which include mainly the topography and environmental features, the stay time, number of men, horizontal and vertical mobility, and sample analysis or return [45]. Recent studies have applied high-resolution remote sensing observation data to assess whether the potential sites meet the above requirements, especially in the topography and environmental features [15,32,52–55]. The successful Chang'e missions considered both the technical feasibility of the mission and the realization of the scientific objectives in the selection of the landing site. The site selection process was informed by a set of widely recognized criteria, which included

more than 4 engineering constraints and 7 scientific constraints [39] (Table S1 online).

The selection criteria for this study were based on the technical criteria recognized during the site selection process for the Artemis project and the Chang'e mission. In addition, the specific needs were considered for water ice and resource utilization for future lunar research. To establish criteria for landing site selection, we compiled a comprehensive spatial database that includes terrain mapping, average solar illumination, and average Earth visibility data, the lunar impact crater distribution map, the average, max and min regolith temperature maps in summer and winter, the M3 Ice Exposures map, the PSRs map, the Water Equivalent Hydrogen (WEH) abundance map, the CO₂ cold traps map, the Unified Geologic Map, and the compositional maps of FeO, high-calcium pyroxene, low-calcium pyroxene, olivine, and plagioclase (Table S2 online). The selection of lunar landing sites necessitates consideration of various factors, with safety and scientific value being paramount. Our methodology defines landing feasibility probability (V-lfp) as the dependent factor, influenced by 20 key quantified factors and 3 qualitative factors (Table S3 online). The details of the spatial datasets and the selection factors are described in the Supplementary material Section 1.

2.2. The LFP-based site selection procedure

Fig. 1 illustrates the methodology utilized for lunar landing site selection, employing GA to specify the LFP model, a comprehensive index we developed to evaluate the viability of potential landing sites. The selection process comprises three phases: (1) Compilation of a training dataset that integrates expert insights into landing site selection criteria; (2) Collaborative cross-regional training of the GA-based LFP model; and (3) Application of the LFP model to identify suitable landing sites, considering spatial and morphological constraints. Initially, we convened authorities with extensive experience in China's lunar landing missions, including experts in probe landing safety assessment, lunar water ice radar detection, and lunar geological exploration, to assess the landing areas identified by the USA Artemis program. These experts first defined criteria for landing site selection, considering scientific value and engineering safety, combined with the needs of lunar frontier research in different fields to designate the valuable areas as viable (positive samples) and the hazardous areas as non-viable (negative samples) for landing, thus establishing the training and validation datasets grounded in expert judgment. Subsequently, we defined an objective function based on the discrepancy between the theoretically modeled LFP and the expert-selected sample areas, employing GA to extrapolate the experts' site selection criteria. Pre-training of the model provides an initial assessment of the relative importance of the site selection factors in the expert site selection decision process, and in this way, selecting the most important factors reduces the detrimental effect of factor correlation on the accuracy of the model. Formal training will further optimize the LFP model, and this optimization yielded an LFP map for the lunar south pole region. In the final stage, areas where the LFP and the factors simultaneously satisfy the thresholds trained by GA were classified as potential landing sites. We then segmented the LFP map to isolate areas that satisfy specific landing prerequisites: a minimum size of 2 km², the eccentricity of the external ellipse of the landing area is no greater than 0.9 (the closer the eccentricity is to 1, the flatter the shape of the landing area), and the largest contiguous area comprising at least 30% of the region. Based on engineering requirements for landing and surface exploration, a minimum area of 2 km² is necessary to ensure both landing safety and sufficient space for scientific investigation. Besides, the minimum 2 km² area requirement ensures that selected landing sites are not isolated pixels but instead form contiguous patches. This

criterion helps filter out fragmented or disconnected areas, ensuring that the LFP model identifies viable, consolidated zones rather than scattered individual pixels. By executing these steps, we pinpointed promising landing locations at the lunar south pole. The details of LFP model and mathematical expressions are described in the Supplementary material Section 2.

This selection approach is designed to provide expert teams with a preliminary map of potential landing sites, thereby enhancing and complementing traditional selection methods. The LFP model integrates quantifiable spatial factors to prioritize potential landing sites, while also allowing the incorporation of constraint layers that reflect qualitative assessments. The spatial factors numerically rank each pixel, while the constraint layers function similarly to visual interpretation methods by filtering out sites that fail to meet specific conditions.

In lunar landing and roving exploration, the site selection process should be tailored to different mission scenarios by customizing the factor system and selecting appropriate training samples. The LFP model framework proposed accommodates such flexibility: both the selection factors and training samples can be adjusted to suit different mission types and target regions. As a result, the model can be optimized to identify more purposeful and scenario-specific landing sites, demonstrating its potential for theoretical scalability. To demonstrate the effectiveness of the proposed approach, the factor system used in our experiment references factors from the Chang'e and Artemis mission series, including water ice detection, mineral composition, and geologic diversity. The resulting site selection outputs are therefore best suited for future lunar south pole missions targeting water ice exploration. Because variables such as distance to PSRs, hydrogen content, and proximity to water ice deposits are included, the training samples emphasize pixels suitable for traversing into shadowed regions to detect volatiles. Consequently, water ice detection factors receive higher weights in the optimized LFP model.

3. Results

3.1. The valuable landing sites at the lunar south pole

Here we generated an LFP model that integrates the ESI with the EVI to assess the viability of each pixel for landing. The trained LFP model and assessment are described in the Supplementary material Section 3. This resulted in a comprehensive map of potential landing sites at the lunar south pole, delineating 120 sites across three categories: 25 high-priority, 64 medium-priority, and 31 low-priority locations (Fig. 2). The 25 high-priority sites are strategically situated in the center of 2 km² flat areas, offering abundant safe space for landers touching the lunar. These sites are in close proximity to scientifically valuable craters, within 1 km of the nearest water ice deposits, PSRs, and cold traps, thus offering prime opportunities for water ice detection and geological research. The ideal rover paths extend from these landing sites toward prospective water ice locations and around the PSR perimeters. Furthermore, the deployment of Unmanned Aerial Vehicles (UAVs) could facilitate exploration and sampling within the PSRs themselves. These landing sites adhere to stricter criteria regarding slope, illumination, and regolith temperature, yielding greater accuracy and engineering safety compared to suggested sites over a broad >200 km² area. Notably, 9 high-priority sites, 4 medium-priority sites, and 3 low-priority sites are proximal to proposed landing areas for the USA Artemis III program, cross-validating the value of the selected landing sites. Three of the high-priority sites we identified (①, ②, and ③) are particularly noteworthy. Located centrally with respect to the lunar south pole's geographic

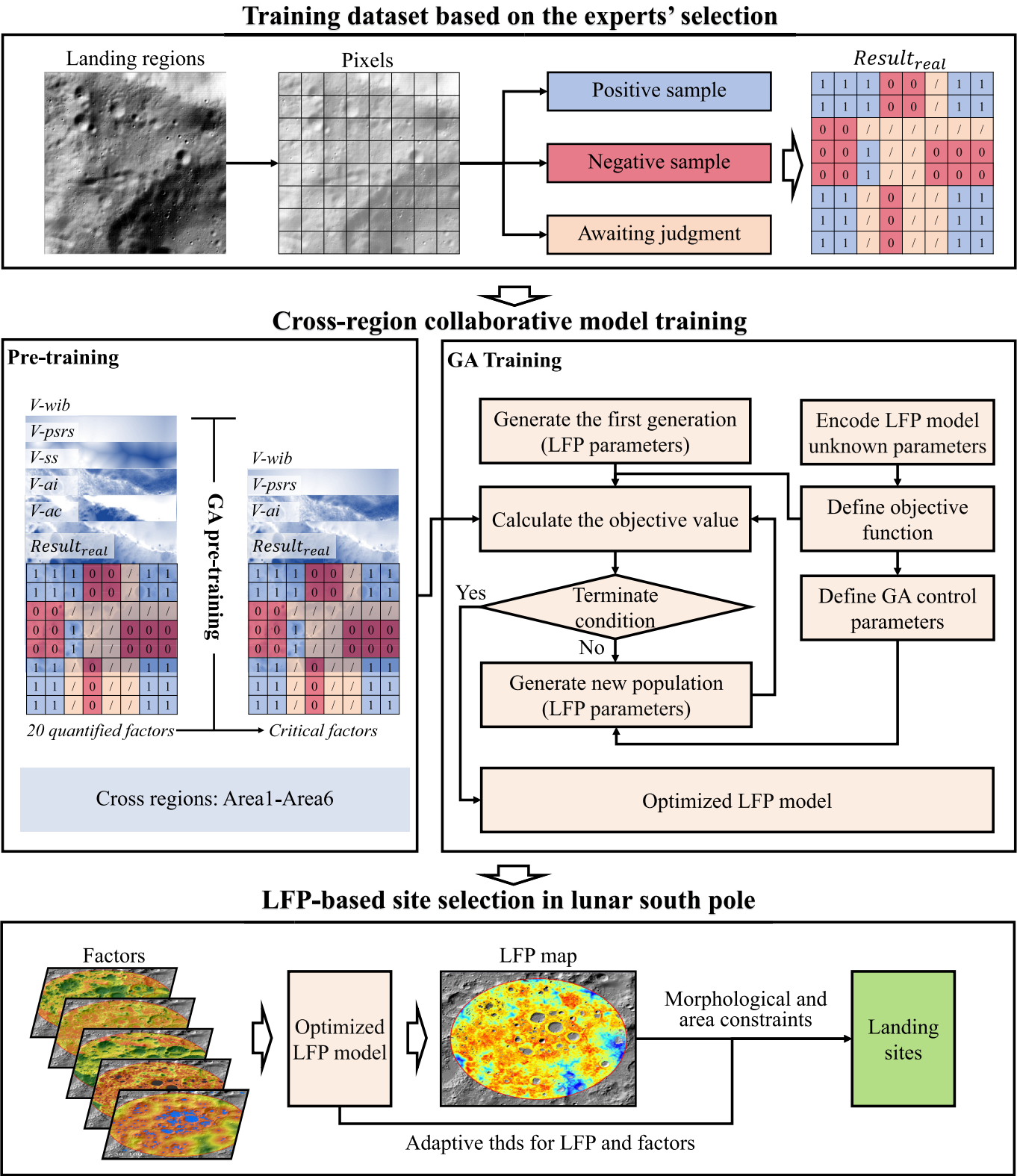


Fig. 1. The workflow of the site selection of the lunar south pole LFP based.

coordinates, these sites are adjacent to the enigmatic Shackleton impact crater and Haworth impact crater, deemed two of the Moon's most intriguing features due to their scientific value, and are also favorably positioned for solar energy collection, which is crucial for engineering safety and mission sustainability [56].

3.2. Landing feasibility probability map

The LFP map (Fig. 3a) validates our site selection methodology. We found that the most favorable landing locations, indicated by the highest LFPs, are situated in proximity to craters, often at the

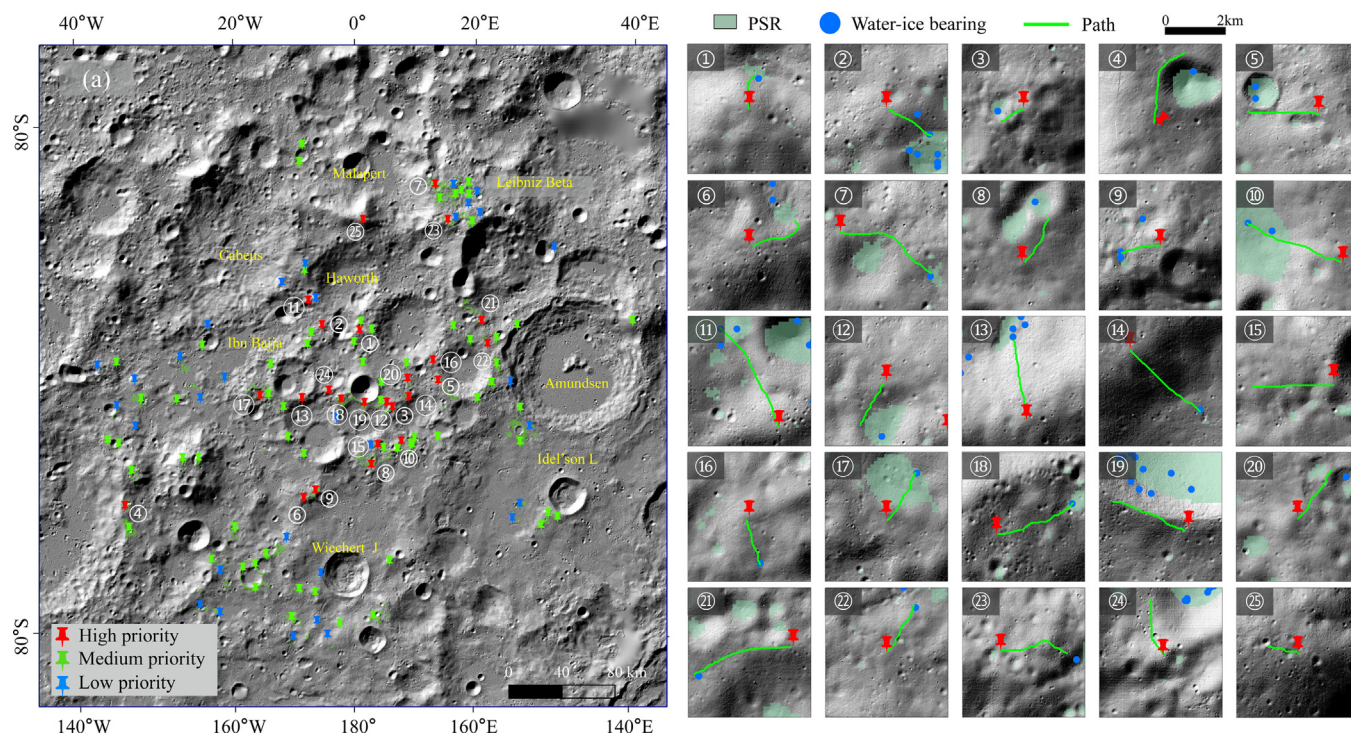


Fig. 2. A detailed map of potential landing sites at the lunar south pole (82°S–90°S). The outcome of our LFP-based analysis identifies 120 candidate sites categorized by priority: 25 high-priority sites, 64 medium-priority sites, and 31 low-priority sites. The high-priority sites, numbered ① to ②⑤, are further illustrated with schematics detailing their precise locations, topographic characteristics, and their respective distances from PSRs and known water ice resources. Notably, high-priority sites ③, ⑤, ⑫, ⑬, ⑯, ⑰, ⑲, ⑳, ㉑, ㉒, ㉓ locate in landing areas of the USA Artemis III program.

crater's edge or in the inter-crater connecting areas. As a result, the predominant patterns of landing sites are circular and linear, while large square landing sites are located in the transitional areas between plateaus and craters. In Fig. 3a, all 120 identified sites exhibit high LFPs ranging from 0.816 to 0.840. Based on LFP and the performance in the 23 selection factors, those deemed sites classified as a high priority have an LFP greater than 0.83, the medium-priority sites have an LFP greater than 0.82, and the low-priority sites have an LFP greater than 0.81. Across the lunar south pole, the LFP distribution approximates a Gaussian curve, with a notable concentration of values around 0.75. The majority of pixels with elevations below 0 m also demonstrate an LFP close to 0.75 (Fig. 3b and c). Similarly, the most common slope gradients are below 7 degrees with corresponding LFP values around 0.75 (Fig. 3b and d). The analysis of illumination conditions revealed that high-priority landing sites exhibit an arithmetic mean illumination value exceeding 0.30 (Fig. 3b and e), indicating favorable illumination stability for sustained operations. Furthermore, LFP scores showed a strong negative correlation with proximity to water ice (Fig. 3b and f), suggesting that regions closer to potential ice deposits were prioritized due to their higher scientific and resource utilization value. These findings align with mission objectives targeting volatile-rich zones. These observations underscore the challenge of pinpointing high-priority landing sites within the lunar south pole. Consequently, the 120 landing sites identified are exceptionally small in size, encompassing a mere 0.6% (551 km²) of the lunar south pole, underscoring their value for lunar exploration.

4. Discussion

4.1. Feasibility validation of landing sites

(1) Explanation analysis of the selection result. The generalized additive model (GAM) was employed to associate each critical

factor with potential lunar landing sites, which were coded binarily (0 representing non-sites, 1 representing sites). This model accounted for 72.3% of the deviance explained by all 10 quantified factors and the p -value of each factor is less than 2×10^{-16} , indicating robust predictive power for this type of binary outcome. The pink areas around the curves in Fig. 4 represent the uncertainty in the GAM model's estimated smooth relationships, specifically the 95% confidence intervals. This shaded region indicates where the true relationship between the factor (e.g., slope or illumination) and the response LFP is likely to lie, with 95% probability. A wider band suggests higher uncertainty, often due to sparse data or high variability in that terrain region, while a narrow band reflects greater confidence in the trend. From an engineering safety standpoint, the slope of the terrain plays a significant role; a slope within the range of 0° to 6.5° positively influences the site selection, suggesting that such sites are within the preferred terrain safety areas (Fig. 4a). The exploration value of a landing site enhances with an increase in average illuminance. Sites with an average illuminance greater than 0.32 demonstrate a strong positive influence on their status as high-value landing sites (Fig. 4b). The exploration value of a landing site is enhanced with an increase in average illuminance. Sites with regolith temperatures between 69 and 125 K demonstrate a strong positive influence on their status as high-value landing sites (Fig. 4c). With regard to scientific goals, the LFP of the true landing sites chosen through our model showed a negative correlation with the distance to water ice deposits (Fig. 4d). Specifically, the desirability of a site for exploration purposes diminishes markedly as the distance from the water ice increases from 0 to 11.5 km. Similarly, the proximity to PSRs inversely affects the selection likelihood for a site, albeit with a less pronounced impact on LFP than the proximity to water ice resources (Fig. 4e). The proximity from the landing site to the cold trap has been observed to exert an increasing and then decreasing influence on the LFP. This positive effect has been noted to intensify for proximity of less than 2180 meters, after which it

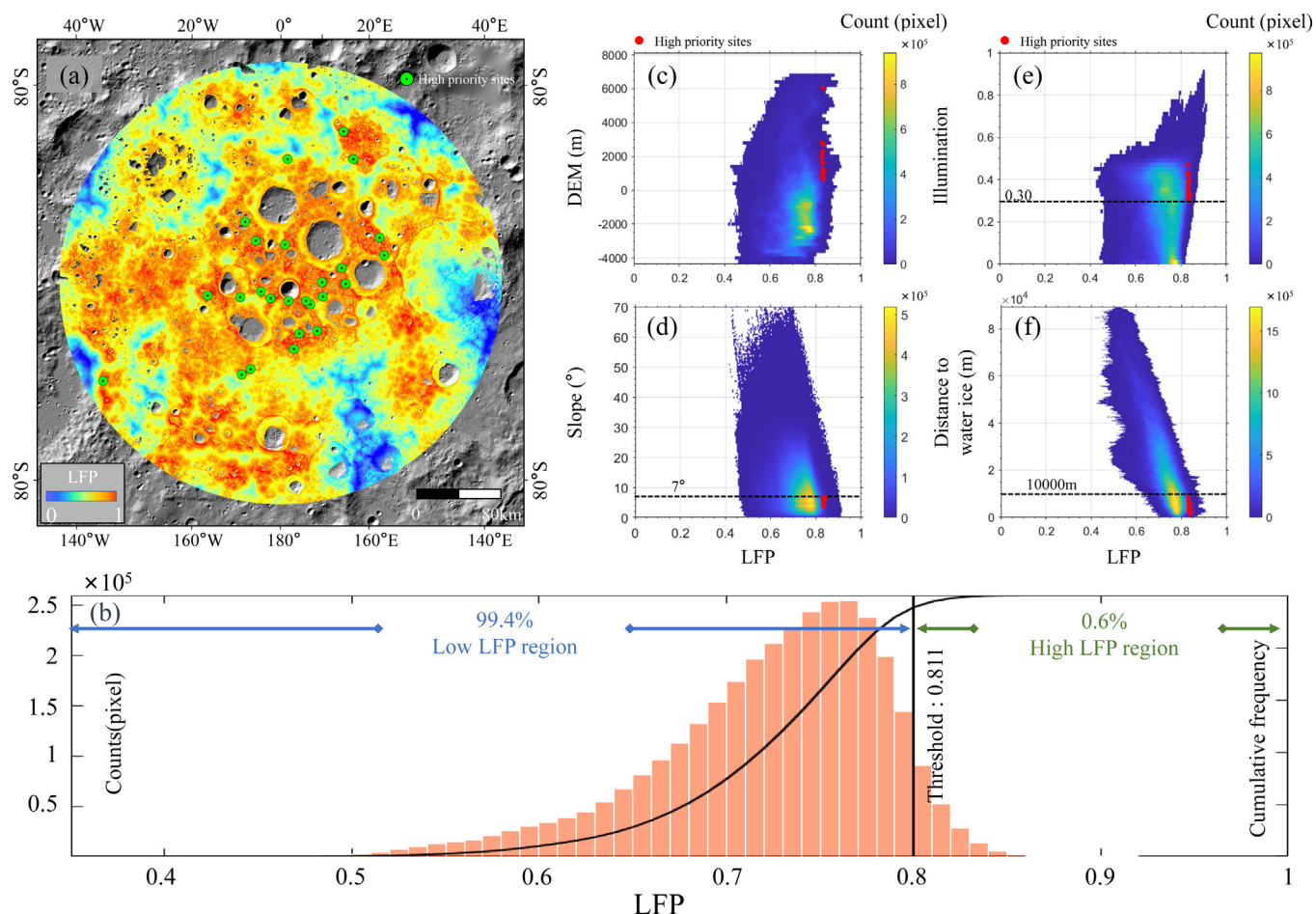


Fig. 3. The LFP map proves the validity of landing sites. (a) The LFP map. (b) The histogram of the LFP that shows 0.6% of the south pole is in prioritization. (c) The relationship between elevation and LFP. (d) The relationship between slope and LFP. (e) The relationship between illumination and LFP. (f) The relationship between distance to water ice and LFP.

appears to diminish (Fig. 4f). The optimal level of geological diversity at the designated landing site is below 0.019 or up 0.031 (Fig. 4g). Furthermore, the mineral abundance of the landing site contributes positively to the LFP, particularly FeO, Pyroxene, and Plagioclase, which play a pivotal role in reversing the lunar history (Fig. 4h, i and j). In GAM analysis, the Effective Degrees of Freedom (EDF) for all 10 site selection factors are greater than 2, suggesting each factor has a strong nonlinear influence on site selection outcomes. Notably, the average illumination and proximity to the water ice exhibited the highest EDF values, both exceeding 8.0, underscoring the necessity of employing a nonlinear LFP model to capture their impact accurately.

(2) Physical criteria validation. A validation of ten critical criteria—3 engineering safety factors (slope, illumination, and regolith temperature) and 7 critical scientific value factors (proximity to water ice, proximity to PSRs, proximity to cold traps, geodiversity, FeO abundance, contribution of the Pyroxene abundance and the Plagioclase abundance, as detailed in Table S6 online)—reveals that all 16 high-priority landing sites satisfy each of the 10 scientific and engineering criteria, and half of them satisfy 9 criteria. In the medium-priority sites, 63 of them meet between 8 and 10 of these criteria, while all low-priority sites fulfill at least 8 criteria.

Among the 120 lunar south pole landing sites identified by our LFP model, the Shackleton Connection Ridge is particularly promising due to its potential for year-round continuous illumination. The Shackleton crater features a permanently shadowed interior, yet M3 spectral analysis indicates possible water ice presence. Thus,

sampling this area could provide vital information about water on the Moon. Both Area 1 and Area 2 in the connecting ridge region offer optimal engineering safety and valuable scientific opportunities (Fig. 5a). The LROC images show no large craters in these areas, which are flat enough for landers (Fig. 5b). Area 1 has an average slope of 3.72°, with 93.2% of its area under 7°, while Area 2 has a slope of 3.46%, with 93.7% under 7°. Both meet slope criteria for lunar landings and are located in the rare high-illumination region of the lunar south pole, ensuring continuous solar energy availability (Fig. 5c). The percentage of areas with Earth visibility greater than 0.4 reached 87.95 percent in Area 1 and 91.44 percent in Area 2, both of which have terrain conditions for continuous communication with the ground (Fig. 5d and e).

4.2. Comparison with other landing sites

To further validate the correctness of the proposed landing site map, the top 5 proposed landing sites were compared with the candidate areas from other studies, including the four landing sites at the Shackleton connection ridge proposed by Sathyan, et al. [36], and the top 4 landing sites selected using the CNN-based model by Feng et al. [15]. Sathyan et al. [36] employed a multidimensional spatial analysis approach to systematically evaluate four sites in the Shackleton connection ridge for the water ice exploration. They prioritized the landing sites based on the expected scientific outcomes and feasibility of accessing a PSR for volatile detection and quantification. Key evaluation metrics include slope, illumination,

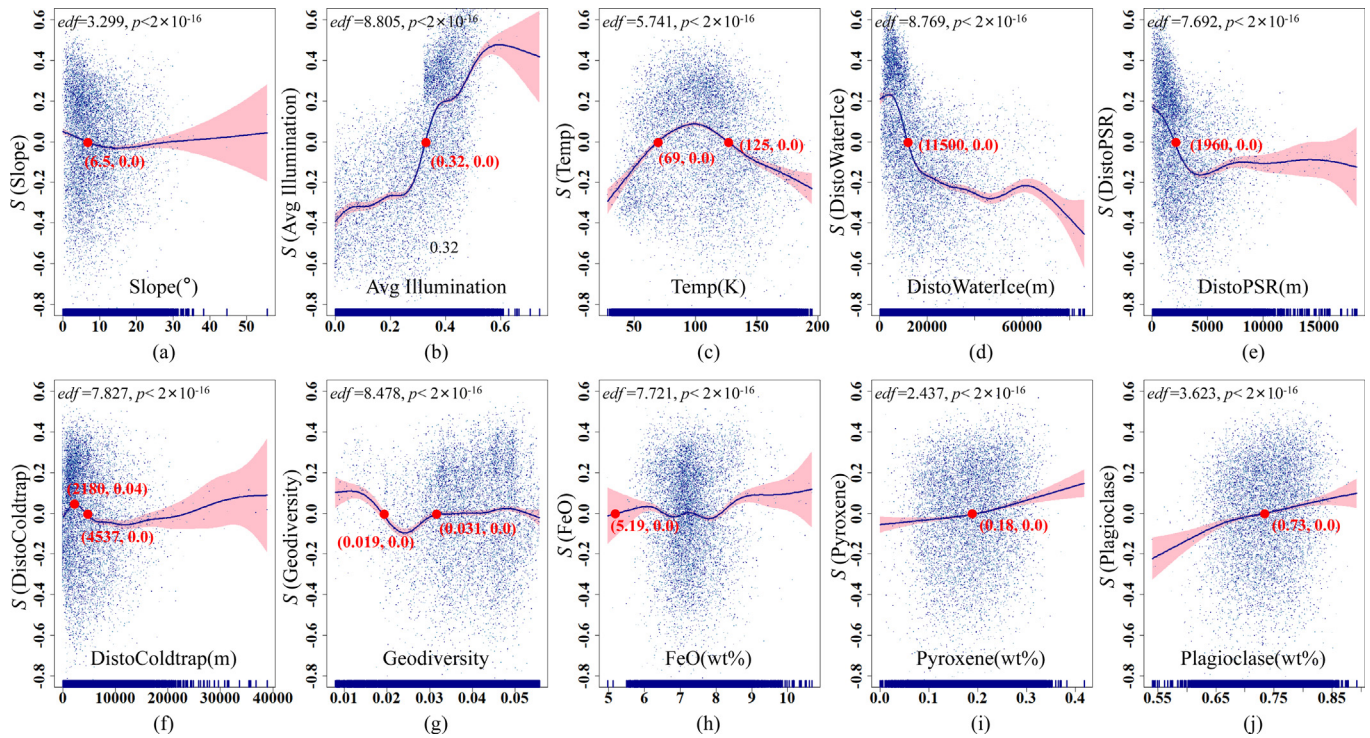


Fig. 4. Relationships between the landing site selection and the influencing factors. (a) The contribution of the surface slope. (b) The contribution of the average illumination. (c) The contribution of the regolith temperature. (d) The contribution of the proximity to the water ice. (e) The contribution of the proximity to the PSR. (f) The contribution of the proximity to the cold traps. (g) The contribution of the Geodiversity. (h) The contribution of the FeO abundance. (i) The contribution of the Pyroxene abundance. (j) The contribution of the Plagioclase abundance.

surface temperature, and accessibility to nearby PSRs. Although refined site selection areas have been proposed, achieving consensus among experts still requires considerable time because of the complexity and multidisciplinary nature of the evaluation process. Instead of directly comparing the relative importance of individual factors, the LFP model leverages expert-selected sample regions to simulate a set of weights and thresholds. This approach helps generate reference weights that support and inform expert decision-making. The 1D-CNN learning approach selected landing groups distributed throughout the lunar south pole, while the mechanism and optimization process were not totally revealed. The proposed LFP method explicitly defines its objective function as minimizing the discrepancy between predicted results and ground-truth data, ensuring full transparency in the optimization process.

Table 1 presents the numerical statistical results evaluating the performance of the sites selected by the proposed LFP method, the sites recommended by Sathyan et al. [36] and Feng et al. [15], based on the 10 critical factors in this study. Bold values highlight instances where a site demonstrates superior performance in a given factor. The analysis indicates that the LFP-selected sites (Sites 1–5) perform particularly well in engineering safety metrics, especially average slope (all below 6.422), outperforming most reference sites in the regolith temperature, which is consistently lower, suggesting more favorable thermal conditions.

Notably, Site 4 excels in proximity to water ice and PSR and the FeO abundance, making it a strong candidate for missions prioritizing resource utilization. Site 3 offers the highest illumination among LFP selections, critical for solar-powered operations. While the reference sites G21 and C2 show advantages in specific factors, like water ice proximity or illumination, the LFP method's selections provide a balanced trade-off across multiple criteria, avoiding weaknesses in any single metric. This is enabled by a comprehensive factor system that ensures more balanced performance across all evaluated dimensions. In contrast, the sites selected by the

CNN-based method show exceptional performance in isolated indicators but underperform in key parameters such as PSR accessibility. Compared to these, the LFP approach offers a more robust solution for identifying landing sites that meet diverse and complex mission requirements without compromising on critical objectives.

4.3. Cluster distribution of landing sites

These landing sites, with their specific coordinates provided, represent the optimal locations for human exploration at the lunar south pole. When prioritizing according to scientific goals, the distribution of the three priority levels of landing sites shows a pronounced clustering around major impact craters and plateaus (Fig. 6a). Notable locations include Shackleton Crater, de Gerlache Crater, Sverdrup West, Faustini Crater, Kuhn Crater, Leibniz Beta Plateau, Amundsen Crater, and Idelson's Crater. These form 7 distinctive clusters, encompassing 52% of the 125 sites identified. The permanently shadowed Shackleton Crater, which maintains internal temperatures consistently below the water ice sublimation threshold of 110 K, hosts 7 high-priority sites, ranging from 3.0 to 7.0 km from water ice, highlighting the Shackleton cluster's significant potential for water ice harvesting and lunar base construction. The de Gerlache Crater serves as a natural lunar laboratory, ideal for studying regolith and harnessing frozen volatiles, potentially providing resources for fuel and construction materials for future base camps. The Sverdrup West cluster, with its low surface slope variability, includes 10 selected landing sites, each with FeO abundances exceeding 5%, thereby meeting the fundamental requirements for human or robotic habitation. The Faustini Crater, with sites surrounding multiple craters, is prime for collecting diverse lunar soil samples. By analyzing the size-frequency distributions of the craters in these areas, researchers can refine impact flux chronology curves and gain insights into the Moon's thermal

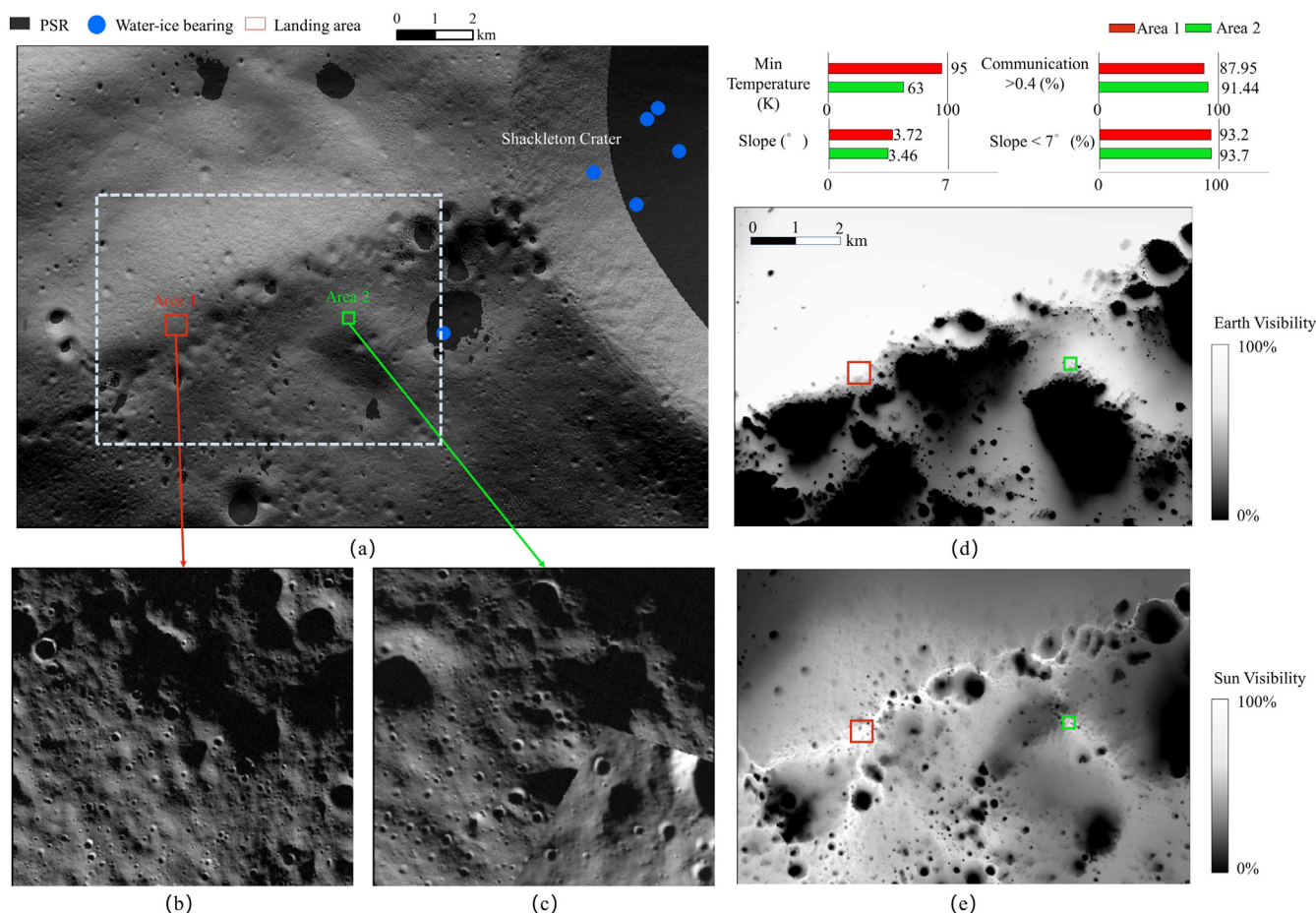


Fig. 5. The landing sites selected of the LFP model in the Shackleton connecting ridge. (a) The locations of landing sites on the hillshade map. (b) The LROC image of Area 1. (c) The LROC image of Area 2. (d) The earth-visibility map. (e) The average illumination map.

history. The Kuhn cluster's landing site, situated at a lower latitude and featuring flatter terrain, is well-suited for installing seismographic equipment for moonquake studies. The Leibniz Beta Plateau cluster, characterized by elevated terrain devoid of mountainous obstructions, boasts optimal illumination and communication conditions, making it an ideal candidate for large-scale bases and lunar astronomical observation experiments due to its high density of landing sites. The Amundsen cluster, located in the South Pole-Aitken (SPA) basin, offers a unique opportunity to address fundamental scientific questions regarding the SPA basin's structure, the Moon's early evolution, and the bombardment history of the inner solar system through strategic sample collection. Lastly, at the Idelson'l crater, 3 out of 5 sites are classified as medium-priority, with distances from 1.1 to 1.8 km to the nearest PSR. This proximity enables comprehensive surveys within a 2 km radius, offering valuable insights into these shadowed regions.

The identified landing sites offer strategic locations not only for constructing lunar bases but also for developing a network of lunar-based observatories (Fig. 6b), creating a crucial platform for humans to monitor Earth and delve into deep space observations. In such a network, the maximum distance between any 2 adjacent sites is a manageable 25 km, less than the 35 km traversed by the Apollo 17 lunar rover [57], thus enabling efficient information exchange, goods transportation, and resource sharing among neighboring sites. A minimum of 4 landing sites within a lunar landing site network is necessary to facilitate autonomous observational missions and ensure mutual support capabilities. Seven

networks satisfy these scientific and engineering prerequisites, including Shackleton, Sverdrup West, Faustini-Amundsen, Leibniz Beta, de Gerlache, Kuhn, and Idel'son L (Fig. 6b). To expedite the establishment of this lunar observation network, we recommend that nations initiate dialogues and cooperation to deploy missions between stations within the same network. Such international collaboration will be instrumental in constructing a sophisticated, multi-faceted observation network on the Moon, serving a broad spectrum of human scientific endeavors.

Among the 7 lunar landing site networks, the Shackleton network stands out with 7 high-priority sites (Fig. 6b, Table S6 online), manifesting the highest potential for scientific exploration and presenting substantial engineering safety in that, when one site is in danger, the other sites can provide timely assistance. Landing site ③ serves as the central hub of this network and is a prospective landing area for the USA Artemis program. Within a network, any 2 landing sites that are within 25 km of each other and maintain line-of-sight can effectively interact and support one another. By this measure, the Shackleton network, with its 26 observational pairs or linkages, presents numerous opportunities for establishing a cohesive network of bases in the region. The Sverdrup West network owns 10 landing sites but numerous 90 connecting lines, forming a dense network for communication and material exchange. The Faustini-Amundsen network, perched on a ridge between the Faustini and Amundsen craters, offers excellent lighting and communication for deploying drones into the craters to gather soil samples and exploit lunar regolith. Its 13 landing sites

Table 1
Comparison between the landing sites selected by the LFP model and the sites selected by Sathyan et al. [36], and the sites selected using CNN by Feng et al. [15].

Name	Evaluation criteria									
	Engineering safety			Scientific value						
	Average Slope	Average Illumination	Regolith temperature	Proximity to water ice	Proximity to PSR	Proximity to CO2 cold traps	Geodiversity	FeO abundance	Pyroxene abundance	Plagioclase abundance
1	5.960	0.385	69.031	2217.072	1841.258	2363.318	0.053	6.705	0.107	0.800
2	6.395	0.351	87.953	2477.132	1152.727	1375.090	0.047	6.757	0.141	0.771
3	6.422	0.457	112.675	3780.805	1843.061	2632.257	0.047	6.656	0.141	0.770
4	6.350	0.365	91.150	1694.824	656.485	922.675	0.031	7.657	0.198	0.704
5	6.113	0.377	88.702	5426.123	1355.227	3038.328	0.054	6.973	0.152	0.757
(Sathyan et al. [36])-S	19.500	0.417	130.070	3070.072	2666.767	2037.028	0.050	6.282	0.057	0.833
(Sathyan et al. [36])-C1	11.373	0.424	131.751	8192.136	1844.286	5729.732	0.050	6.106	0.108	0.815
(Sathyan et al. [36])-C2	7.858	0.514	125.032	8266.735	2086.934	2177.276	0.051	6.490	0.056	0.794
(Sathyan et al. [36])-D	14.670	0.361	121.082	11308.197	2302.414	4409.643	0.051	6.609	0.150	0.783
(Feng et al. [15])-G5	5.795	0.486	120.067	2580.021	7821.711	5020.141	0.036	0.738	0.196	0.738
(Feng et al. [15])-G14	6.488	0.396	120.275	2488.905	6843.236	4563.795	0.738	6.1389	0.086	0.805
(Feng et al. [15])-G15	7.809	0.559	110.101	2071.696	4414.987	2939.063	0.047	6.646	0.136	0.768
(Feng et al. [15])-G21	5.036	0.459	124.7179	644.752	64921.783	1146.85	0.042	6.170	0.171	0.808

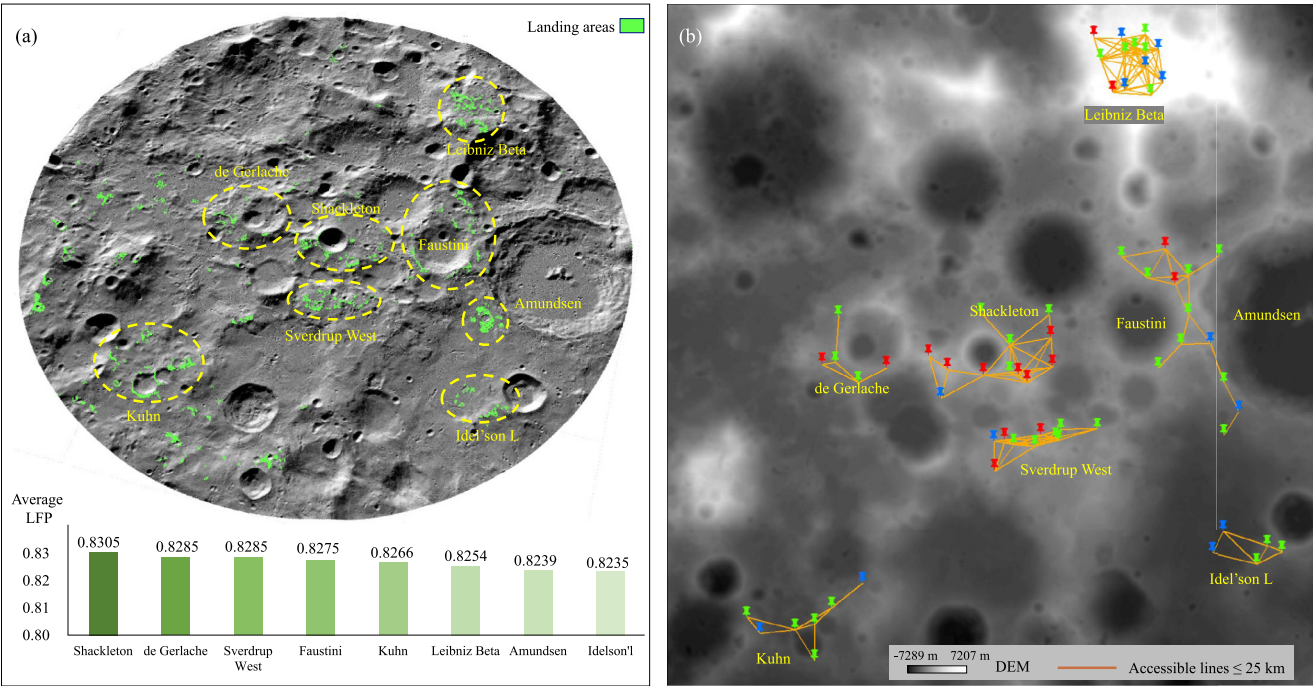


Fig. 6. Cluster distribution of major landing sites and their spatial connectivity. (a) 8 exploration clusters for science goals. (b) 7 landing site networks for the construction of lunar bases.

and 36 connecting lines create a unique, scoop-shaped network, providing exceptional conditions for both lunar-based astronomical observations and moonquake studies. The Leibniz Beta network, situated on the lunar highland, benefits from unimpeded views over lunar mountain ranges and is particularly conducive to moon-based astronomical observations. Each site in this network boasts an average of 9.07 visual connections to its neighbors, signaling a high density of sites and connectivity, which is ideal for a robust lunar observational network. The de Gerlache network, despite having 2 fewer stations than the Kuhn network, boasts 8 additional lines, reflecting a denser and more evenly distributed network configuration. The de Gerlache, Kuhn, and Idel'son L networks each contain fewer than 10 sites; among them, the de Gerlache network, with the highest density of only 2.57, indicates that this area is well-suited for compact lunar base networks.

4.4. The assessment of the LFP model

In the pursuit of identifying lunar landing sites with significant scientific value and superior engineering safety for forthcoming lunar missions, this study introduces the LFP model. The model is designed to assess landing feasibility, employing a GA with RMSE serving as the objective function to extract site selection heuristics and align with expert knowledge. The LFP model, as trained, demonstrates an F1 score exceeding 94% on the training dataset, affirming its alignment with expert site selection experience. The level of accuracy is commendable, yet it hinges on several assumptions: (1) the assumed candidate landing sites are inherently optimal for safe touchdowns and offer maximal scientific benefits. This assumption has been substantiated for engineering safety, and high scientific value is anticipated, and (2) it presumes the com-

plete exclusion of areas unsuitable for landing, and its modeling residuals only indicate the exclusion of unfeasible landing sites, but the feasibility for the landing of certain areas far from water ice but with extremely flat topography is not yet determined.

While expert-provided physical thresholds and the AHP method contribute to a rigorous selection process [49]. However, the different disciplinary backgrounds of the experts within the site selection workgroup can result in different factor weights, which increases the cost of time and the difficulty of site selection. The LFP model avoids this problem by not requiring experts to compare the relative importance of factors, but rather outputs the uniform factor weights oriented to a specific mission scenario using the suitable and unsuitable landing samples identified by all experts. Besides, the GA heuristically determines factor weights and optimizes model parameters, diminishing the impact of subjective biases and preventing the emergence of inconsistent site selection criteria. Furthermore, the LFP model recognizes that acceptable landing sites should not only possess high LFP scores but also meet specific thresholds for each contributing factor. These thresholds are not arbitrary fixed values; they are optimal combinations derived from the training samples through the GA. This approach ensures that selected sites maintain high feasibility and strong performance across all factors, showcasing the LFP model's capability for multi-objective optimization in the context of lunar landing site selection.

The LFP-based 120 landing sites at the lunar south pole streamline the selection process for future lunar exploration, saving time and effort for lunar science experts and engineers. This allows scientists to concentrate on assessing the detailed scientific potential of each site while also providing a thorough evaluation of both the landing safety and the feasibility of long-term exploration.

5. Conclusion

Considering the intricate conditions of the lunar south pole and the indeterminate scientific values it holds, the dual challenges of ensuring engineering safety and achieving scientific goals, the criteria for landing in this region are stringent. To tackle these two challenges, a collaborative heuristic GA was developed and applied across different areas. The LFP concept, based on the ESI and the EVI, was introduced and utilized to enable the identification of potential landing sites. We generated the LFP map, indicating that a mere 0.6% of the Moon's south pole area, encompassing 551 square kilometers, qualifies as viable landing terrain that meets the dual criteria of engineering safety and scientific value. Within this expanse, there are 120 valuable potential landing sites, categorized into 25 high-priority, 64 medium-priority, and 31 low-priority targets. The methodology was tested on prospective landing sites for the USA Artemis program, demonstrating that the F1 Score of landing site selection for the datasets exceeds 94.59%. The model boasts a low RMSE of 0.0773, signifying an extremely low mis-selection or wrong-selection rate of landing areas, testifying to the robustness and reliability of the method.

The methodology and tools developed in this study represent a significant advancement from the traditional expert-driven site selection process, which has underpinned decades of human deep space exploration. We have transitioned to a heuristic intelligent optimization approach, capable of large-scale landing site determination that incorporates expert knowledge into discriminative rule learning. This versatile method can be extended beyond the Moon, potentially streamlining site selection for missions targeting Mars, Jupiter, Mercury, and various asteroids. The 120 identified sites and the 7 interconnected networks we have established facilitate a more thorough investigation into the vast scientific potential of lunar south pole exploration. They also lay the ground-

work for a foundational network concept that supports international collaboration in lunar endeavors. Consequently, this work accelerates the progress of human exploration of the Moon, setting the stage for future discoveries and international space exploration initiatives.

Conflict of interest

The authors declare that they have no conflict of interest.

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Author contributions

Xiaohua Tong conceived and oversaw the study. Yongjiu Feng designed the experiment and wrote the manuscript. Pengshuo Li performed the experiments and wrote the manuscript. Haoteng Li assisted with the data production. Yan Liu provided support for the revision of the manuscript. Mengrong Xi supported additional data processing. Rong Wang provided advice on the methodology. Shurui Chen, Panli Tang, and Yuze Cao provided the temporal data. Qian Huang, Qionghua You, Shijie Liu, Zhen Ye, and Yusheng Xu provided the valuable lunar terrain and illumination map. Xiong Xu, Chao Wang, Yanmin Jin, and Sicong Liu gave feedback on the analyses and the manuscript. Huan Xie and Huaiyu He provided ideas and coordinated the project. All the authors participated in manuscript preparation.

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.scib.2025.08.035>.

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